MIL-O-41838D
6 January 1978
SUPERSEDING
MIL-O-41838C
30 May 1974

MILITARY SPECIFICATION

OVERSHOES, WOMEN'S, PLASTIC, GRAY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a vinyl plastic women's overshoe.

1.2 Classification. The overshoe shall be of one type and width only, and of the following sizes as specified (see 6.2):

Sizes - 4, 5, 6, 7, 8, 9, 10, 11, 12

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-553	-	Paper, Wrapping, Tissue
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-676	-	Boxes, Setup

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8435

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STANDARDS

FEDERAL

FED-STD-406 - Plastics: Methods of Testing
FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Samples.

3.1.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample in which case this specification shall govern.

3.1.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.4).

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3.2 Material.

* 3.2.1 Vinyl. The vinyl molding compound shall consist of vinyl resins, plasticizers, and stabilizers which are suitable for slush molding overshoes which conform to the requirements contained in this specification.

* 3.2.2 Button assembly. The button assembly shall be the manufacturer's commercial standard. If the button assembly shown in section A-A of figure 1 is used, a washer shall be required. The color of the buttons shall be clear, black, or approximately match the color of the overshoe.

3.2.3 Fastening loop. The "figure 8" fastening loop shall consist of a nylon braid with an elastic core and a corrosion resistant metal clip suitable for forming around the braid without fracturing. The fastening loop shall be of such dimension as to assure proper closure of the overshoe. The small end of the "figure 8" fastening loop shall have a maximum diameter of 3/16 inch (5 mm) to assure retention on the button. The fastening loop shall be attached by looping the short end over the top button.

3.2.3.1 Nylon braid. The nylon braid shall be a woven casing of nylon yarn and shall be 0.095 ± 0.010 inch (2.41 ± 0.25 mm) in diameter. The braid shall have a minimum breaking strength of 18.0 pounds (80 N).

3.2.3.2 Loop clip. The clip shall be constructed of brass or steel of a hardness and thickness suitable for complete forming around the braid without fracturing. Steel clips, if used, shall be plated with nickel or coated with gray enamel.

* 3.3 Design. The overshoe shall be of the pull-on, vinyl plastic boot type with a button and loop fastening closure (see figure 1). The overall outside surface shall be of a texture or grain design which creates a matte or dull finish. The sole tread shall be the contractor's design. The top contour may be straight or slightly curved. The tread design shall be no more than 50 percent of the total thickness in any area of the sole.

3.3.1 Color. The color of the finished vinyl overshoe shall be gray approximately matching color No. 36118 of FED-STD-595 or the color may be smoke.

3.4 Molds. The molds shall be provided by the contractor.

3.5 Construction.

3.5.1 The overshoe shall be slush molded using a vinyl chloride plastisol.

3.5.1.1 Physical properties. The overshoe shall conform to the physical requirements specified in table I when tested as specified in 4.4.4.

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TABLE I. Overshoe requirements

Characteristics	Requirements
Tensile strength, minimum	1000 psi (6.9 MPa)
Elongation at break, minimum	225 percent
Tensile stress (modulus) at 100 percent elongation, minimum	320 psi (2.2 MPa)
Tear resistance, minimum	100 pounds per inch of thickness (17.5 kN/m)
Durometer, hardness, maximum	67

3.5.2 Thickness. The thickness of the finished overshoe shall conform to the requirements of table II.

TABLE II. Overshoe thickness

Location	Minimum thickness
Upper (1 inch (25 mm) down from top, center of side)	0.025 inches (0.64 mm)
Heel (directly above sole binding, in upper area)	0.050 inches (1.27 mm)
Sole (toe)	0.055 inches (1.40 mm)
Sole (ball)	0.055 inches (1.40 mm)
Sole (extreme heel)	0.125 inches (3.18 mm)

3.5.3 Dimensions of overshoe. The dimensions (measurements and contours) shall conform to the measurements as detailed in figure 2 for the size 7 overshoe. The height measurement for the size 7 overshoe shall be between 7 and 7-1/2 inches (178 and 190 mm) measured at the back of overshoe from the inside heel seat upwards to the top back center. For all other sizes, grading shall be employed as detailed in figure 2.

3.5.4 Cracking. The overshoe sole shall show no evidence of cracking when tested as specified in 4.4.4.

3.5.5 Leakage. The overshoe shall show no evidence of leaking when tested as specified in 4.4.4.

3.5.6 Finish. The completed overshoe shall be coated on the inside with a slip finish for easy donning and doffing. The outer surface shall be sprayed with a clear vinyl acrylic lacquer so as to produce a matte or dull appearance. The overshoe shall not have any tacky or sticky areas.

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3.5.7 Button placement. One button shall be placed on the inner overshoe side [illegible] and at least 1/2 inch (13 mm) from the [illegible] of the front towards the outside from [illegible]. One button shall be placed on the outer overshoe side approximately [illegible] forepart of the loop over the buttons shall create a neat lap-over of the forward section, of the overshoe over the instep. The button positions shall be marked for each size according to the best commercial grading practice.

3.5.8 Replacement of defective components. During the manufacturing process, components [illegible] material defects or damages that are classified as defects in 4.4.2.1 and 4.4.2.2 shall be removed from production and replaced with non-defective and properly matched components.

3.6 Marking and identification. The size, in numerals 1/4 to 1/2 inch (6 to 13 mm) in height, shall be embossed or branded in the shank area. In addition, the contractor's name or identification symbol shall be embossed or branded in the shank area with characters not less than 1/4 inch (6 mm) in height. As an alternate, the contractor's name or identification symbol may be printed with black or white water resistant ink and shall be located on the inside of the upper forepart below the button fastener loop. The height of the characters shall be as specified above.

3.7 Workmanship. The finished overshoe shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 6.2), the first article submitted in accordance with 3.1.2, shall be visually inspected as specified in 4.4.2 for compliance with design, construction, workmanship, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

* 4.4.1.1 Certification. The contractor shall furnish a certificate of compliance for the components listed below:

Component	Characteristic	Requirement paragraph
Braid	Material identification	3.2.3.1
	Breaking strength	3.2.3.1
	Diameter	3.2.3.1
Loop clip	Material identification	3.2.3.2
	Coating (when required)	3.2.3.2

4.4.2 Examination of the end item. The defects found during examination shall be classified in accordance with 4.4.2.1 and 4.4.2.2. The inspection levels and acceptable quality levels shall be as indicated in 4.4.2.3.

4.4.2.1 Visual examination of the end item. The overshoes shall be examined for the defects listed below. The sample unit shall be one completely fabricated overshoe and the selection shall be by pairs. The lot size shall be expressed in units of one overshoe. For pairing examination, the pair shall be examined together. Each defect found during the examination for pairing shall be scored as a single defect.

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Examine	Defect
[illegible]	[illegible] not right and [illegible] of same size Variation in color between pair Variation in height over 1/4 inch (6 mm)
[illegible]	Any spot or stain seriously affecting appearance Any embedded foreign matter not readily removed without leaving a thin spot
Design	Any characteristic not in accordance with specified requirements Not one piece without seams
Color and finish (applicable to all components)	Color not uniform or not as specified Outside finish not as specified Finish tacky or sticky
Loop	Cut or frayed Clip missing, mispositioned, or not securely attached
Vinyl	Any hole (including pinhole), cut, tear, rupture, or crack Any blister, weak or thin spot, or bubble, affecting serviceability
Construction and workmanship	Any part of button malformed, broken, missing, or otherwise defective affecting serviceability Any part of button assembly not securely affixed Overshoe malformed
Marking, identification and size	Missing, illegible, misplaced, or incorrect

* 4.4.2.1 Dimensional examination. The overshoe shall be examined for dimensional defects to determine conformance to requirements specified in 3.3, 3.5.2, 3.5.3, and 3.5.7. The thickness shall be measured using a dial micrometer with a throat at least 12 inches (305 mm) deep. The anvils of the micrometer shall be 3/8 inch (10 mm) diameter. Any dimension that is not within the established tolerances shall be classified as a defect.

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* 4.4.2.3 Inspection levels [illegible] inspection levels and AQLs [illegible] follows:

Examination	Inspection level	AQL
4.4.2.1	II	6.5
4.4.2.2	S-3	6.5

* 4.4.3 Packaging inspection. An examination shall be made to determine that the preservation-packaging, packing, and marking comply with the section requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Examination for the closure defects listed below shall be made on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be level S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking exterior and interior	Omitted; incorrect, illegible; of improper size, location, sequence or method of application
Material	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling Bulged or distorted container
Content	Number of interior packages per shipping container is more or less than required Size indicated on overshoes not the same as specified on packages or shipping container 1/

1/ For this defect, one package from each container in the sample shall be examined.

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4.4.4 End item testing. The finished overshoe shall be tested for the characteristics indicated in table III. The sample unit shall be one overshoe for cracking and leakage tests. The sample unit for all other tests shall be two overshoes and the results shall be reported as the average of the three values obtained from each sample unit. The lot size shall be expressed in units of one overshoe. Requirements are applicable to the sample unit. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. End item tests

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as
Tensile strength	3.5.1.1	4.5.1	3	Nearest 10 psi (0.1 MPa)
Elongation at break	3.5.1.1	4.5.1	3	Nearest 1 percent
Tensile stress (modulus) at 100 percent elongation	3.5.1.1	4.5.1	3	Nearest 10 psi (0.1 MPa)
Tear resistance	3.5.1.1	4.5.2	3	Nearest 1 pound per inch of thickness (0.1 kN/m)
Durometer hardness	3.5.1.1	4.5.3	3	Nearest whole number
Cracking	3.5.4	4.5.4	1	Pass or fail
Leakage	3.5.5	4.5.5	1	Pass or fail

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4.5 Test methods.

4.5.1 Physical determinations. The tensile strength, elongation at break and tensile stress (modulus) at 100 percent elongation shall be determined on the same specimens using Method 1131 of Fed-STD-406. The following exceptions shall apply:

The sample length shall be 4-3/4 inches (121 mm), the width in the narrowest portion shall be 0.25 inch (6.40 mm), the gauge length shall be 1 inch (25.4 mm) and the testing instrument shall operate at 12 inches per minute (305 mm/min).

4.5.2 Tear resistance. The tear resistance shall be run as specified in Method 1131 of FED-STD-406.

4.5.3 Durometer hardness. The durometer hardness shall be measured as specified in Method 1082 of FED-STD-406.

4.5.4 Cracking test. The overshoe sole shall be folded across the ball from side to side and then folded back so that the edges are together. A drop of acetone shall be ejected from a dropper onto the fold ridge and allowed to stand for approximately 5 seconds. The sole shall be examined for evidence of cracking.

4.5.5 Leakage test. Each overshoe to be tested shall be immersed in water to within 1/2 ± 1/8 inch (13 ± 3 mm) of the lower fastening button. The immersed overshoe shall remain in the water for a sufficient length of time to insure that no leaks occur when the inside of the overshoe is examined for visible penetration of water. Examination for water penetration may be made by the use of a suitable mechanical or electronic device.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each pair of overshoes shall be preserved-packaged in a snug-fitting paperboard box conforming to PPP-B-566 or PPP-B-676. Each box shall be closed in accordance with the appendix of the applicable box specification. Prior to packaging, each overshoe shall be separated from the other and from the inside surfaces of the box with at least one thickness of tissue paper conforming to type I, class 1 of UU-P-553.

5.1.2 Level C. Overshoes shall be preserved-packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A packing. Twenty-four pair of overshoes of one size only, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2 Level B packing. Twenty-four pair of overshoes of one size only, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3. Toward the end of the contract or when there are less than the required amount of the same size, mixed sizes may be packed within the same container.

* 5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636. The container shall be assembled and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3.

* 5.2.3 Level C packing. Overshoes, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches (130 by 100 mm) with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

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6. NOTES

6.1 Intended use. The overshoes are intended for use by military personnel of the Department of [illegible].

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) When a first article is required (see 3.1.2, 4.3, and [illegible]).
(d) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Sample. For access to samples, address the procuring activity issuing the invitation for bids.

6.4 First article. When a first article sample is required; it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one pair of overshoes in any size. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for inspection and approval of the first article.

6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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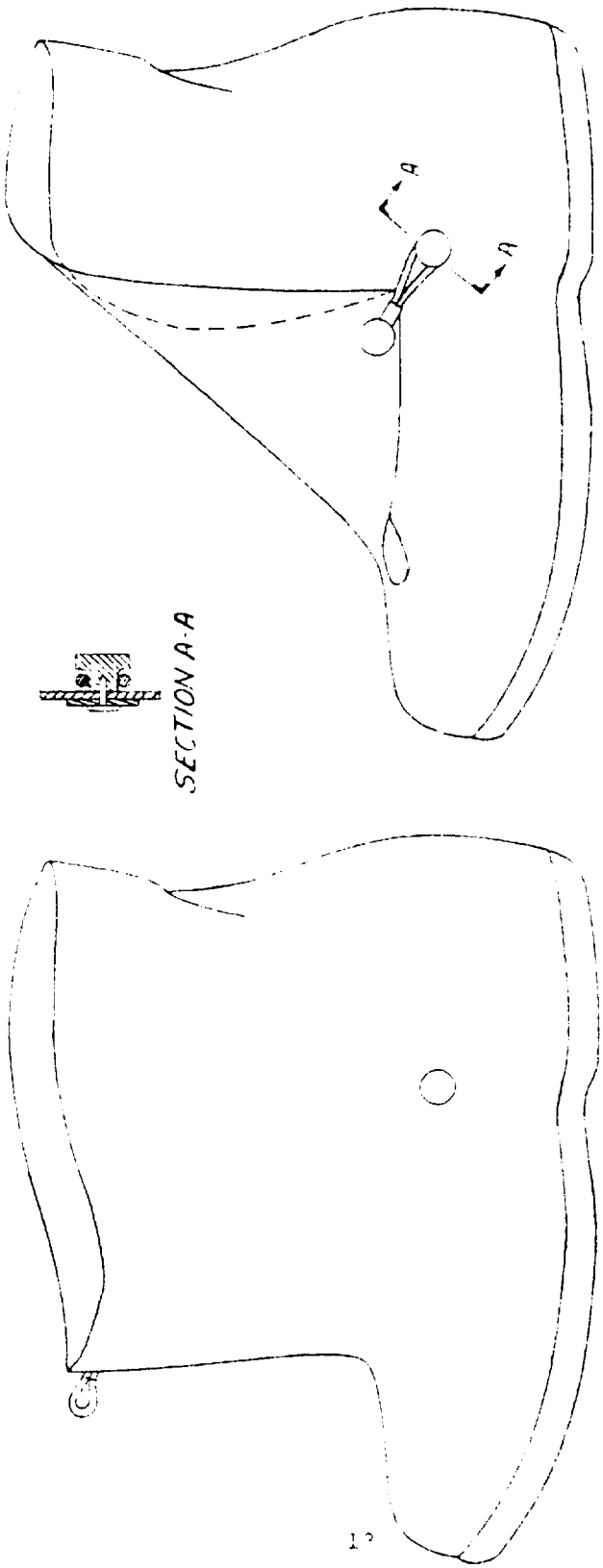


FIG. 1 - OVERSHOE, WOMEN'S

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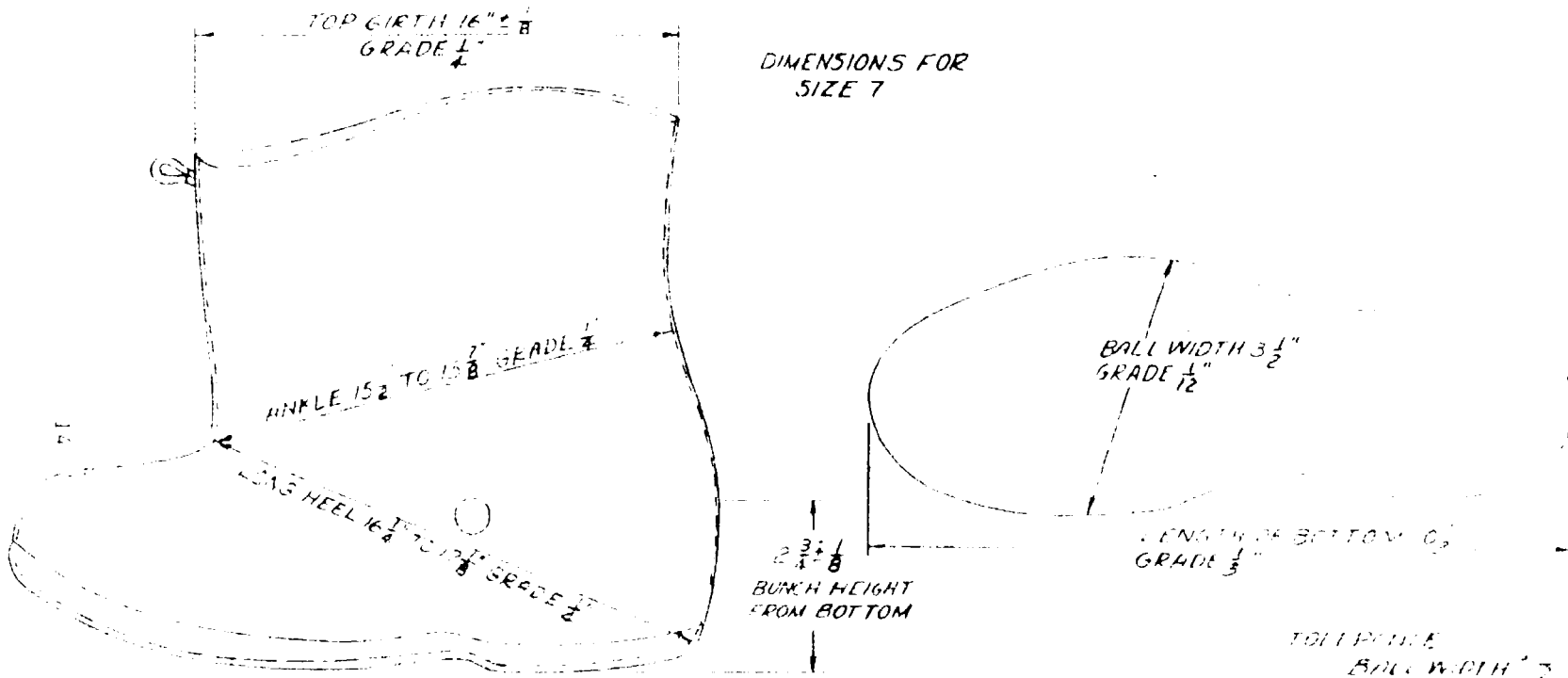


FIG 2. OVERSHOE, WOMEN'S

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